Ganita Manjari — Grade 9 (Part II)

Chapter 14: Math of Space (Surface Area and Volume) — Exercise Set 14.3 Solutions

Question 1

> **Find the total surface area of a cone, if its slant height is 21 m and diameter of its base is 24 m.**

Easy Step-by-Step Solution:

Given: Slant height $l = 21\ m$, Diameter = 24 m ⇒ Radius $r = 12\ m$.

Total Surface Area (TSA) = $\pi r(l + r)$

$$TSA = (22/7) \times 12 \times (21 + 12) = (22/7) \times 12 \times 33 = 8712 / 7 \approx 1244.57\ m^2$$

Final Answer: 1244.57 m² (or 8712/7 m²)

Question 2

> **Find the curved surface area of a right circular cone whose slant height is 10 cm and base radius is 7 cm.**

Easy Step-by-Step Solution:

Given: $l = 10\ cm$, $r = 7\ cm$.

Curved Surface Area (CSA) = πrl

$$CSA = (22/7) \times 7 \times 10 = 220\ cm^2$$

Final Answer: 220 cm²

Question 3

The height of a cone is 16 cm, and its base radius is 12 cm. Find the curved surface area and the total surface area of the cone. (Use π = 3.14)

Easy Step-by-Step Solution:

Given: $h = 16\ cm$, $r = 12\ cm$.

Step 1: Find slant height (l)

$$l = \sqrt{(r^2 + h^2)} = \sqrt{(12^2 + 16^2)} = \sqrt{(144 + 256)} = \sqrt{400} = 20\ cm$$

Step 2: Curved Surface Area (CSA)

$$CSA = \pi rl = 3.14 \times 12 \times 20 = 753.6\ cm^2$$

Step 3: Total Surface Area (TSA)

$$TSA = \pi r(l + r) = 3.14 \times 12 \times (20 + 12) = 37.68 \times 32 = 1205.76\ cm^2$$

Final Answers: CSA = 753.6 cm² | TSA = 1205.76 cm²

Question 4

A cone has a height of 15 cm. If its volume is 1570 cm³, find the radius of the base. (Use π = 3.14)

Easy Step-by-Step Solution:

Volume of Cone = $(1/3)\pi r^2 h$

$$1570 = (1/3) \times 3.14 \times r^2 \times 15$$

$$1570 = 3.14 \times 5 \times r^2 = 15.7\ r^2$$

$$r^2 = 1570 / 15.7 = 100 \Rightarrow r = \sqrt{100} = 10\ cm$$

Final Answer: 10 cm

Question 5

The curved surface area of a cone is 308 cm² and its slant height is 14 cm. Find (i) radius of the base, (ii) total surface area of the cone.

Easy Step-by-Step Solution:

(i) Radius of base:

$$CSA = \pi rl = 308 \Rightarrow (22/7) \times r \times 14 = 308 \Rightarrow 44r = 308 \Rightarrow r = 7\ cm$$

(ii) Total surface area:

$$TSA = CSA + Base\ Area = 308 + \pi r^2 = 308 + (22/7)\times 7^2 = 308 + 154 = 462\ cm^2$$

Final Answers: (i) 7 cm | (ii) 462 cm²

Question 6

A joker's cap is in the form of a right circular cone of base radius 7 cm and height 24 cm. Find the area of the sheet required to make 10 such caps.

Easy Step-by-Step Solution:

Given: $r = 7\ cm$, $h = 24\ cm$.

Slant height $l = \sqrt{(7^2 + 24^2)} = \sqrt{(49 + 576)} = \sqrt{625} = 25\ cm$

CSA of 1 cap = $\pi rl = (22/7) \times 7 \times 25 = 550\ cm^2$

Area of sheet for 10 caps = $10 \times 550 = 5500\ cm^2$

Final Answer: 5500 cm²

Question 7

What length of tarpaulin 3 m wide is required to make a conical tent of height 8 m and base radius 6 m? Assume extra material required for margins and wastage is 20 cm = 0.2 m. (Use π = 3.14)

Easy Step-by-Step Solution:

Given: $h = 8\ m$, $r = 6\ m$.

Slant height $l = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10\ m$

CSA of tent = $\pi rl = 3.14 \times 6 \times 10 = 188.4\ m^2$

Required length of tarpaulin without wastage = $Area / Width = 188.4 / 3 = 62.8\ m$

Total length required including wastage = $62.8\ m + 0.2\ m = 63\ m$

Final Answer: 63 m

Question 8

A right triangle with sides 6 cm, 8 cm and 10 cm is rotated through 360° about the side of 8 cm. Find the volume and the curved surface area of the solid so formed.

Easy Step-by-Step Solution:

Rotating around the 8 cm side forms a cone where:

Radius $r = 6\ cm$, Height $h = 8\ cm$, Slant height $l = 10\ cm$.

Volume: $V = (1/3)\pi r^2 h = (1/3) \times (22/7) \times 6^2 \times 8 = (1/3) \times (22/7) \times 36 \times 8 = 301.71\ cm^3$ $(or\ 96\pi\ cm^3)$

Curved Surface Area: $CSA = \pi rl = (22/7) \times 6 \times 10 = 188.57\ cm^2$ $(or\ 60\pi\ cm^2)$

Final Answers: Volume = 96π cm³ (301.71 cm³) | CSA = 60π cm² (188.57 cm²)

Question 9

Suppose you have a cup in the shape of a right circular cone. Fill it with water to half the depth of the cone. What fraction of the volume of the cup is occupied by the water?

Easy Step-by-Step Solution:

Let full height = H and full radius = R.

Water height = $h = H / 2$. By similar triangles, water radius = $r = R / 2$.

Full volume of cup = $V_{full} = (1/3)\pi R^2 H$

Volume of water = $V_{water} = (1/3)\pi (R/2)^2 (H/2) = (1/3)\pi (R^2/4) (H/2) = (1/8) \times [(1/3)\pi R^2 H] = (1/8) V_{full}$

Final Answer: 1 / 8 of the total volume